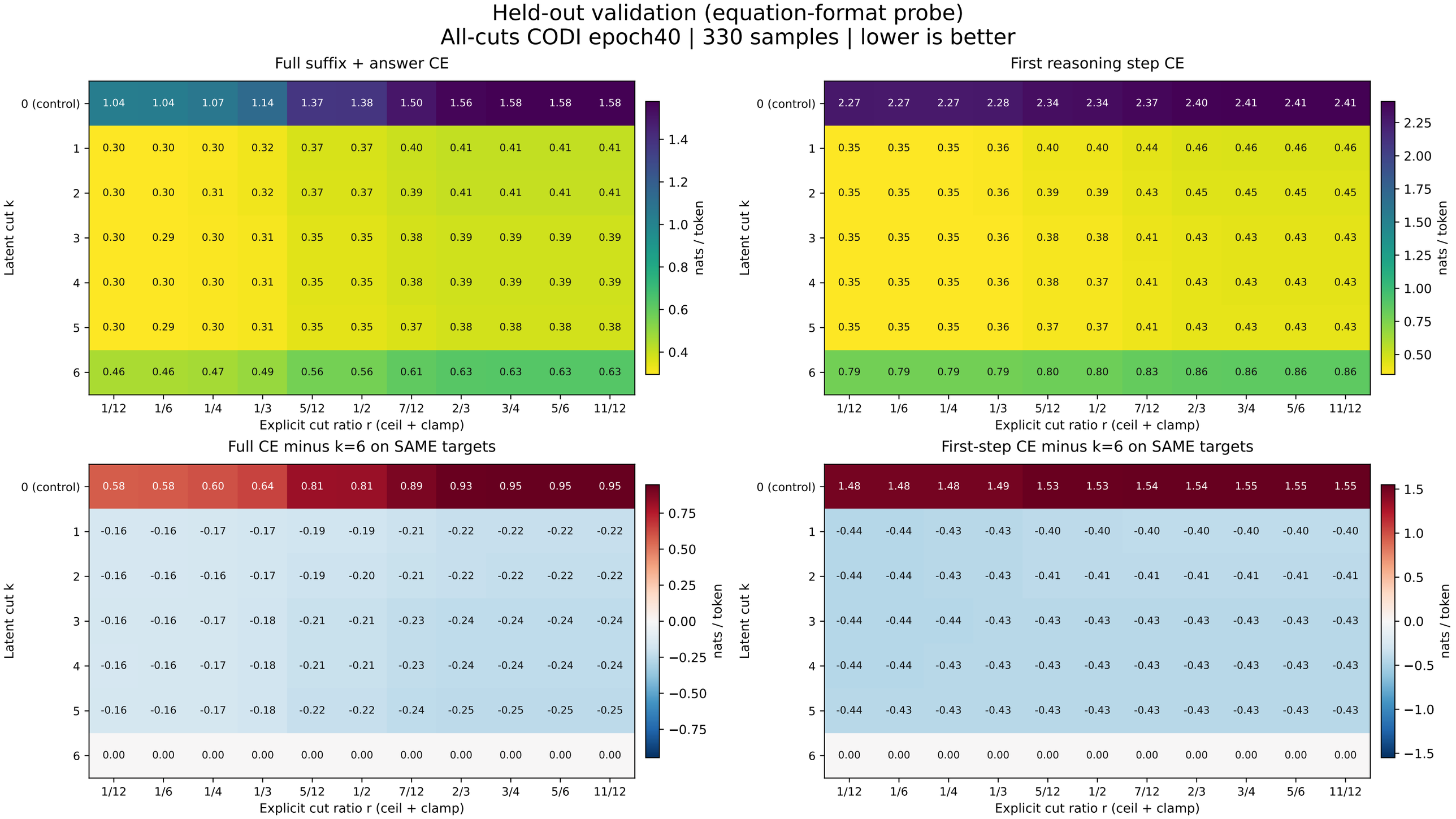
Held-out validation (equation-format probe)
All-cuts CODI epoch40 | 330 samples | lower is better
Full suffix + answer CE
First reasoning step CE
Full CE minus k=6 on SAME targets
First-step CE minus k=6 on SAME targets
Latent cut k
Explicit cut ratio r (ceil + clamp)
1/12 1/6 1/4 1/3 5/12 1/2 7/12 2/3 3/4 5/6 11/12
0 (control) 1.04 1.04 1.07 1.14 1.37 1.38 1.50 1.56 1.58 1.58 1.58
1 0.30 0.30 0.30 0.32 0.37 0.37 0.40 0.41 0.41 0.41 0.41
2 0.30 0.30 0.31 0.32 0.37 0.37 0.39 0.41 0.41 0.41 0.41
3 0.30 0.29 0.30 0.31 0.35 0.35 0.38 0.39 0.39 0.39 0.39
4 0.30 0.30 0.30 0.31 0.35 0.35 0.38 0.39 0.39 0.39 0.39
5 0.30 0.29 0.30 0.31 0.35 0.35 0.37 0.38 0.38 0.38 0.38
6 0.46 0.46 0.47 0.49 0.56 0.56 0.61 0.63 0.63 0.63 0.63
nats / token
1.4 1.2 1.0 0.8 0.6 0.4
0 (control) 2.27 2.27 2.27 2.28 2.34 2.34 2.37 2.40 2.41 2.41 2.41
1 0.35 0.35 0.35 0.36 0.40 0.40 0.44 0.46 0.46 0.46 0.46
2 0.35 0.35 0.35 0.36 0.39 0.39 0.43 0.45 0.45 0.45 0.45
3 0.35 0.35 0.35 0.36 0.38 0.38 0.41 0.43 0.43 0.43 0.43
4 0.35 0.35 0.35 0.36 0.38 0.37 0.41 0.43 0.43 0.43 0.43
5 0.35 0.35 0.35 0.36 0.37 0.37 0.41 0.43 0.43 0.43 0.43
6 0.79 0.79 0.79 0.79 0.80 0.80 0.83 0.86 0.86 0.86 0.86
nats / token
2.25 2.00 1.75 1.50 1.25 1.00 0.75 0.50
0 (control) 0.58 0.58 0.60 0.64 0.81 0.81 0.89 0.93 0.95 0.95 0.95
1 -0.16 -0.16 -0.17 -0.17 -0.19 -0.19 -0.21 -0.22 -0.22 -0.22 -0.22
2 -0.16 -0.16 -0.16 -0.17 -0.19 -0.20 -0.21 -0.22 -0.22 -0.22 -0.22
3 -0.16 -0.16 -0.17 -0.18 -0.21 -0.21 -0.23 -0.24 -0.24 -0.24 -0.24
4 -0.16 -0.16 -0.17 -0.18 -0.21 -0.21 -0.23 -0.24 -0.24 -0.24 -0.24
5 -0.16 -0.16 -0.17 -0.18 -0.22 -0.22 -0.24 -0.25 -0.25 -0.25 -0.25
6 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
nats / token
0.75 0.50 0.25 0.00 −0.25 −0.50 −0.75
0 (control) 1.48 1.48 1.48 1.49 1.53 1.53 1.54 1.54 1.55 1.55 1.55
1 -0.44 -0.44 -0.43 -0.43 -0.40 -0.40 -0.40 -0.40 -0.40 -0.40 -0.40
2 -0.44 -0.44 -0.43 -0.43 -0.41 -0.41 -0.41 -0.41 -0.41 -0.41 -0.41
3 -0.44 -0.44 -0.44 -0.43 -0.43 -0.43 -0.43 -0.43 -0.43 -0.43 -0.43
4 -0.44 -0.44 -0.43 -0.43 -0.43 -0.43 -0.43 -0.43 -0.43 -0.43 -0.43
5 -0.44 -0.43 -0.43 -0.43 -0.43 -0.43 -0.43 -0.43 -0.43 -0.43 -0.43
6 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
nats / token
1.5 1.0 0.5 0.0 −0.5 −1.0 −1.5